

Detailed Machine Learning Lecture Sheet

Overview

This lecture sheet provides detailed lecture-ready material for the following machine learning topics: probabilistic discriminative models, log-linear models, kernel methods, sparse kernel machines, Bayesian methods, k-nearest neighbors, neural networks, clustering, mixture models, Expectation Maximization, sequential data, and Markov models. Each topic includes concept explanations, equation derivations, advantages, applications, and multiple solved mathematical problems.

Probabilistic Discriminative Models

Basic Idea

A probabilistic discriminative model directly learns the conditional probability $P(y | x)$, instead of modeling the joint distribution $P(x, y)$ or class-conditional density $P(x | y)$. The main objective is to separate classes based on observed features.

Suppose input features are represented by $x \in \mathbb{R}^d$, and output class label is y . For classification, a model assigns a score to each class and converts the scores into probabilities.

General Derivation of Conditional Probability Form

Let the score of class k be:

$$a_k(x) = w_k^T x + b_k$$

These raw scores are not probabilities because:

- They can be negative,
- They do not sum to 1.

To convert them into probabilities, use exponentiation so that values become positive:

$$\exp(a_k(x))$$

Now normalize by the sum over all classes:

Why Discriminative Models Are Important

Discriminative Models

1. Directly classify data into classes
2. Usually give higher classification accuracy
3. Work well with large datasets
4. Can handle complex decision boundaries
5. Focus on boundary between classes
6. Widely used in real-world applications like spam filtering and image recognition

$$P(y = k | x) = \frac{\exp(a_k(x))}{\sum_{j=1}^K \exp(a_j(x))}$$

This gives the softmax function. The result satisfies:

- $P(y = k | x) \geq 0$
- $\sum_k P(y = k | x) = 1$

Why Discriminative Models Are Important

They focus only on the decision boundary, so they often perform better than generative models when there is enough labeled data and the main goal is prediction.

Advantages

- Directly models class probabilities.
- Works well for supervised classification.
- Usually easier to optimize for prediction tasks.
- Often requires fewer assumptions about data generation.

Disadvantages dekha lgbe

Applications

- Disease classification
- Fraud detection
- Sentiment analysis
- Spam filtering

Disadvantages of Discriminative Models

1. Need large amount of labeled data
2. Do not model data distribution
3. Can overfit on small datasets
4. Sensitive to noisy data
5. Training can be computationally expensive
6. Less effective when data is missing or limited

Solved Problem 1

Suppose there are two classes with scores:

$$a_1 = 2, a_2 = 1$$

Find $P(y = 1 | x)$ and $P(y = 2 | x)$.

Solution:

Using softmax:

$$P(y = 1 | x) = \frac{e^2}{e^2 + e^1}$$

Using approximate values $e^2 \approx 7.389$, $e^1 \approx 2.718$:

$$P(y = 1 | x) = \frac{7.389}{7.389 + 2.718} = \frac{7.389}{10.107} \approx 0.731$$

Similarly,

$$P(y = 2 | x) = \frac{2.718}{10.107} \approx 0.269$$

Solved Problem 2

Suppose a three-class classifier gives scores $a_1 = 0$, $a_2 = 2$, $a_3 = 1$. Find the probability of each class.

Solution:

$$e^0 = 1, e^2 \approx 7.389, e^1 \approx 2.718$$

Sum:

$$1 + 7.389 + 2.718 = 11.107$$

So,

$$P(y = 1 | x) = \frac{1}{11.107} \approx 0.090$$

$$P(y = 2 | x) = \frac{7.389}{11.107} \approx 0.665$$

$$P(y = 3 | x) = \frac{2.718}{11.107} \approx 0.245$$

Log-Linear Models

Introduction

A log-linear model assumes that the logarithm of the odds, or the logarithm of class probability ratio, is linear in the input features.

For binary classification:

$$P(y = 1 | x) = \sigma(z) = \frac{1}{1 + e^{-z}}, z = w^T x + b$$

Derivation of Logistic Function from Odds

For binary classification, define:

$$P(y = 1 | x) = p, P(y = 0 | x) = 1 - p$$

The odds of class 1 are:

$$\frac{p}{1 - p}$$

Now assume the log-odds is linear in x :

$$\log \left(\frac{p}{1 - p} \right) = w^T x + b$$

Exponentiate both sides:

$$\frac{p}{1 - p} = e^{w^T x + b}$$

Multiply both sides by $(1 - p)$:

$$p = e^{w^T x + b}(1 - p)$$

Expand:

$$p = e^{w^T x + b} - p e^{w^T x + b}$$

Bring terms containing p to one side:

$$p + p e^{w^T x + b} = e^{w^T x + b}$$

Factor out p :

$$p(1 + e^{w^T x + b}) = e^{w^T x + b}$$

Therefore:

$$p = \frac{e^{w^T x + b}}{1 + e^{w^T x + b}}$$

Multiply numerator and denominator by $e^{-(w^T x + b)}$:

$$p = \frac{1}{1 + e^{-(w^T x + b)}}$$

This is the logistic sigmoid function.

Decision Boundary Derivation

Predict class 1 when:

$$P(y = 1 \mid x) > 0.5$$

Since sigmoid is monotonic increasing:

$$\frac{1}{1 + e^{-z}} > 0.5$$

This implies:

$$z > 0$$

Therefore the decision boundary is:

$$w^T x + b = 0$$

Advantages

- Interpretable coefficients.
- Outputs probabilities.
- Fast optimization.
- Effective baseline for classification.

Applications

- Loan approval prediction
- Customer churn prediction
- Medical risk modeling

Solved Problem 1

Given $w = [2, -1]$, $x = [3, 1]$, $b = -2$, find $P(y = 1 | x)$.

Solution:

$$z = (2)(3) + (-1)(1) - 2 = 3$$

$$P(y = 1 | x) = \frac{1}{1 + e^{-3}} \approx 0.9526$$

Solved Problem 2

Find the decision boundary for $w = [1, 2]$, $b = -6$.

Solution:

Decision boundary is:

$$w^T x + b = 0$$

So:

$$1 \cdot x_1 + 2 \cdot x_2 - 6 = 0$$

Thus,

$$x_1 + 2x_2 = 6$$

Solved Problem 3

For $x = [1, 2]$, $w = [0.5, 1]$, $b = -1$, find the class probability.

Solution:

$$z = 0.5(1) + 1(2) - 1 = 1.5$$

$$P(y = 1 | x) = \frac{1}{1 + e^{-1.5}} \approx 0.8176$$

Kernel Methods

Introduction

Many datasets are not linearly separable in the original input space. Kernel methods solve this by implicitly mapping data into a higher-dimensional space.

Let $\phi(x)$ be a transformation from original space to feature space. Instead of computing $\phi(x)$ explicitly, use a kernel:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Why Kernel Trick Works

Suppose a linear classifier in transformed space is:

$$f(x) = w^T \phi(x) + b$$

Assume w is expressed as a linear combination of transformed training points:

$$w = \sum_{i=1}^N \alpha_i \phi(x_i)$$

Substitute into the classifier:

$$f(x) = \left(\sum_{i=1}^N \alpha_i \phi(x_i) \right)^T \phi(x) + b$$

Using distributive property:

$$f(x) = \sum_{i=1}^N \alpha_i \phi(x_i)^T \phi(x) + b$$

Now replace $\phi(x_i)^T \phi(x)$ by kernel $K(x_i, x)$:

$$f(x) = \sum_{i=1}^N \alpha_i K(x_i, x) + b$$

So the model depends only on kernel evaluations.

Common Kernels

Linear Kernel

$$K(x_i, x_j) = x_i^T x_j$$

Polynomial Kernel

$$K(x_i, x_j) = (x_i^T x_j + c)^d$$

RBF Kernel

$$K(x_i, x_j) = \exp \left(-\frac{\|x_i - x_j\|^2}{2\sigma^2} \right)$$

Advantages

- Handles nonlinearity.
- Elegant use of high-dimensional features.
- Strong performance in classification and regression.

Applications

- Face recognition
- Gene classification
- Fault detection

Solved Problem 1

Let $x_1 = [1, 2]$, $x_2 = [3, 4]$. Compute linear kernel.

Solution:

$$K(x_1, x_2) = 1 \cdot 3 + 2 \cdot 4 = 11$$

Solved Problem 2

Using polynomial kernel with $c = 1$, $d = 2$, compute $K(x_1, x_2)$.

Solution:

$$K(x_1, x_2) = (11 + 1)^2 = 144$$

Solved Problem 3

Let $x_1 = [1, 1]$, $x_2 = [2, 3]$, $\sigma = 1$. Compute the RBF kernel.

Solution:

First compute squared Euclidean distance:

$$\|x_1 - x_2\|^2 = (1 - 2)^2 + (1 - 3)^2 = 1 + 4 = 5$$

Then,

$$K(x_1, x_2) = \exp\left(-\frac{5}{2}\right) \approx 0.0821$$

Sparse Kernel Machines**Concept**

A dense kernel model may use all training points. Sparse kernel machines keep only a subset, reducing cost.

The prediction function is:

$$f(x) = \sum_{i \in S} \alpha_i K(x_i, x) + b$$

where S is the selected sparse subset.

Why Sparsity Appears

In methods such as SVM, only support vectors have nonzero coefficients. If a point does not affect the margin, its coefficient becomes zero.

So if the full expansion is:

$$f(x) = \sum_{i=1}^N \alpha_i K(x_i, x) + b$$

and many $\alpha_i = 0$, then:

$$f(x) = \sum_{i \in S} \alpha_i K(x_i, x) + b$$

This is the sparse form.

Advantages

- Lower memory cost.
- Faster testing.
- Better interpretability of influential points.

Applications

- Support Vector Machines
- Relevance Vector Machines
- Sparse classification systems

Solved Problem 1

Let $\alpha_1 = 2$, $\alpha_2 = -1$, $b = 0.5$, $K(x_1, x) = 3$, $K(x_2, x) = 1$. Find $f(x)$.

Solution:

$$f(x) = 2(3) - 1(1) + 0.5 = 5.5$$

Solved Problem 2

Suppose three training points have coefficients $\alpha = [0, 1.5, 0, -2]$. How many support vectors are there?

Solution:

Only nonzero coefficients count. Nonzero values are 1.5 and -2 .

Answer: 2 support vectors.

Bayesian Methods

Introduction

Bayesian classification is based on Bayes' theorem:

$$P(C_k | x) = \frac{P(x | C_k)P(C_k)}{P(x)}$$

Derivation of Bayes' Theorem

From conditional probability:

$$P(C_k | x) = \frac{P(C_k, x)}{P(x)}$$

Also,

$$P(x | C_k) = \frac{P(C_k, x)}{P(C_k)}$$

Therefore,

$$P(C_k, x) = P(x | C_k)P(C_k)$$

Substitute into the first equation:

$$P(C_k | x) = \frac{P(x | C_k)P(C_k)}{P(x)}$$

This is Bayes' theorem.

Posterior Maximization Rule

For classification, choose the class with maximum posterior:

$$\hat{C} = \arg \max_k P(C_k | x)$$

Since $P(x)$ is the same for all classes, this becomes:

$$\hat{C} = \arg \max_k P(x | C_k)P(C_k)$$

Advantages

- Interpretable uncertainty.
- Incorporates prior beliefs.
- Good for probabilistic reasoning.

Applications

- Medical diagnosis

- Spam filtering
- Decision support systems

Solved Problem 1

Suppose:

- $P(C_1) = 0.6$
- $P(C_2) = 0.4$
- $P(x | C_1) = 0.5$
- $P(x | C_2) = 0.2$

Find $P(C_1 | x)$.

Solution:

$$P(x) = 0.5(0.6) + 0.2(0.4) = 0.38$$

$$P(C_1 | x) = \frac{0.5(0.6)}{0.38} = 0.7895$$

Solved Problem 2

Suppose:

- $P(C_1) = 0.3$
- $P(C_2) = 0.7$
- $P(x | C_1) = 0.8$
- $P(x | C_2) = 0.4$

Find the predicted class.

Solution:

Compare unnormalized posteriors:

$$P(x | C_1)P(C_1) = 0.8(0.3) = 0.24$$

$$P(x | C_2)P(C_2) = 0.4(0.7) = 0.28$$

Since $0.28 > 0.24$, predict C_2 .

k-Nearest Neighbors (k-NN)

Introduction

k-NN stores training data and classifies a new sample based on its nearest examples.

Distance Derivation

For two points in 2D, distance comes from Pythagoras' theorem:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Extending to n -dimensions:

$$d(x, x_i) = \sqrt{\sum_{j=1}^n (x_j - x_{ij})^2}$$

This is Euclidean distance.

Prediction Rule

1. Compute distance from test point to all training points.
2. Sort distances.
3. Select the nearest k neighbors.
4. Use majority vote for classification.

Advantages

- Very easy to use.
- No explicit training phase.
- Naturally handles multiclass problems.

Applications

- Text classification
- Medical decision support

- Recommendation engines

Solved Problem 1

Test point (3,3), training points:

- (2,2), Class 0
- (4,4), Class 1
- (3,5), Class 1

Use $k = 3$.

Solution:

$$d_1 = \sqrt{(3-2)^2 + (3-2)^2} = \sqrt{2}$$

$$d_2 = \sqrt{(3-4)^2 + (3-4)^2} = \sqrt{2}$$

$$d_3 = \sqrt{(3-3)^2 + (3-5)^2} = 2$$

Classes are 0,1,1, so majority is class 1.

Solved Problem 2

Let the training points be:

- (1,1), Class A
- (2,2), Class A
- (5,5), Class B

Classify test point (3,3) with $k = 1$.

Solution:

Distances:

$$d((3,3), (1,1)) = \sqrt{8} \approx 2.828$$

$$d((3,3), (2,2)) = \sqrt{2} \approx 1.414$$

$$d((3,3), (5,5)) = \sqrt{8} \approx 2.828$$

Nearest point is (2,2), Class A.

Answer: Class A.

Solved Problem 3

Using the same data, classify (3,3) with $k = 3$.

Solution:

All three points are considered. Two are class A and one is class B.

Answer: Class A.

Neural Networks

Introduction

A neural network computes weighted sums followed by nonlinear activations.

For one neuron:

$$z = w^T x + b$$

$$a = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Derivation of Neuron Equation

Suppose a neuron receives inputs $x_1, x_2, \dots, x_n$ with weights $w_1, w_2, \dots, w_n$. Each input contribution is multiplied by its weight:

$$w_1 x_1, w_2 x_2, \dots, w_n x_n$$

Sum all contributions and add bias:

$$z = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b$$

In vector form:

$$z = w^T x + b$$

Then apply activation:

$$a = \sigma(z)$$

Why Activation Is Needed

Without activation, multiple layers would collapse into one linear transformation. Nonlinearity allows neural networks to learn complex patterns.

Advantages

- Learns highly nonlinear functions.
- Flexible architectures.
- High predictive power.

Applications

- Computer vision
- NLP
- Speech recognition
- Time-series prediction

Solved Problem 1

Given $w = [1, -2]$, $x = [2, 1]$, $b = 0.5$, find neuron output.

Solution:

$$z = 1(2) + (-2)(1) + 0.5 = 0.5$$

$$a = \frac{1}{1 + e^{-0.5}} \approx 0.6225$$

Solved Problem 2

Given $w = [0.5, 0.5]$, $x = [4, 2]$, $b = -1$, find output with sigmoid.

Solution:

$$z = 0.5(4) + 0.5(2) - 1 = 2 + 1 - 1 = 2$$

$$a = \frac{1}{1 + e^{-2}} \approx 0.8808$$

Solved Problem 3

Suppose a neuron output threshold for class 1 is 0.5. If the neuron output is 0.27, what is the predicted class?

Solution:

Since $0.27 < 0.5$, predict class 0.

Clustering

Introduction

Clustering groups similar objects together without labels.

k-Means Objective Derivation

Suppose there are K clusters with centroids $\mu_1, \mu_2, \dots, \mu_K$. For each point assigned to cluster C_i , measure squared distance to the centroid.

Total objective:

$$J = \sum_{i=1}^K \sum_{x \in C_i} \|x - \mu_i\|^2$$

This is the within-cluster sum of squares.

Derivation of Centroid Update

For one cluster with points $x_1, x_2, \dots, x_n$, minimize:

$$J(\mu) = \sum_{i=1}^n (x_i - \mu)^2$$

Differentiate with respect to μ :

$$\frac{dJ}{d\mu} = \sum_{i=1}^n 2(\mu - x_i) = 2n\mu - 2 \sum_{i=1}^n x_i$$

Set derivative to zero:

$$2n\mu - 2 \sum_{i=1}^n x_i = 0$$

$$n\mu = \sum_{i=1}^n x_i$$

Thus,

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

So the optimal centroid is the mean of cluster points.

Advantages

- Simple and fast.
- Useful for exploratory analysis.
- Scales well for moderate-sized data.

Applications

- Customer segmentation
- Grouping similar documents
- Market analysis

Solved Problem 1

Find centroid of points 2,4,6.

Solution:

$$\mu = \frac{2 + 4 + 6}{3} = 4$$

Solved Problem 2

Points 1,3,5,7 belong to one cluster. Find centroid.

Solution:

$$\mu = \frac{1 + 3 + 5 + 7}{4} = 4$$

Solved Problem 3

If a point $x = 6$ belongs to a cluster with centroid $\mu = 4$, find squared error contribution.

Solution:

$$(x - \mu)^2 = (6 - 4)^2 = 4$$

Mixture Models

Introduction

A mixture model assumes data is drawn from several component distributions.

$$p(x) = \sum_{k=1}^K \pi_k p_k(x)$$

where π_k are mixing coefficients and $\sum_k \pi_k = 1$.

Derivation of Mixture Density

Suppose one hidden variable z indicates which component generated the sample. Then:

$$P(x) = \sum_{k=1}^K P(x, z = k)$$

Using product rule:

$$P(x) = \sum_{k=1}^K P(x | z = k)P(z = k)$$

Let:

- $P(z = k) = \pi_k$

- $P(x | z = k) = p_k(x)$

Then:

$$p(x) = \sum_{k=1}^K \pi_k p_k(x)$$

For Gaussian mixture:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k)$$

Advantages

- Models overlapping clusters.
- Provides soft memberships.
- Useful for probabilistic density estimation.

Applications

- Speech recognition
- Image segmentation
- Customer behavior analysis

Solved Problem 1

Suppose $\pi_1 = 0.7$, $p_1(x) = 0.2$, $\pi_2 = 0.3$, $p_2(x) = 0.5$. Find $p(x)$.

Solution:

$$p(x) = 0.7(0.2) + 0.3(0.5) = 0.29$$

Solved Problem 2

If $\pi_1 = 0.4$, $\pi_2 = 0.6$, $p_1(x) = 0.1$, $p_2(x) = 0.8$, find $p(x)$.

Solution:

$$p(x) = 0.4(0.1) + 0.6(0.8) = 0.04 + 0.48 = 0.52$$

Expectation Maximization Algorithm

Introduction

EM is used when data contains hidden variables, such as component labels in mixture models.

Log-Likelihood Challenge

For data $x_1, \dots, x_N$, the likelihood is:

$$L(\theta) = \prod_{n=1}^N \sum_{k=1}^K \pi_k p(x_n | \theta_k)$$

Taking log gives:

$$\log L(\theta) = \sum_{n=1}^N \log \left(\sum_{k=1}^K \pi_k p(x_n | \theta_k) \right)$$

This is hard to maximize directly because of the log of a sum.

E-Step Derivation

Compute posterior probability that point x_n belongs to component k :

$$\gamma(z_{nk}) = P(z_n = k | x_n)$$

Using Bayes' theorem:

$$\gamma(z_{nk}) = \frac{\pi_k p(x_n | \theta_k)}{\sum_{j=1}^K \pi_j p(x_n | \theta_j)}$$

For Gaussian mixtures:

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_n | \mu_j, \Sigma_j)}$$

M-Step Derivation

Expected effective number of points in cluster k :

$$N_k = \sum_{n=1}^N \gamma(z_{nk})$$

Updated mean is weighted average:

$$\mu_k = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) x_n$$

Updated mixing coefficient:

$$\pi_k = \frac{N_k}{N}$$

Advantages

- Elegant hidden-variable optimization.
- Widely used in mixture models.
- Soft cluster assignments improve flexibility.

Applications

- Gaussian mixture clustering
- Missing data estimation
- Latent variable modeling

Solved Problem 1

Suppose $\pi_1 \mathcal{N}_1 = 0.3$, $\pi_2 \mathcal{N}_2 = 0.2$. Find responsibility of component 1.

Solution:

$$\gamma(z_{n1}) = \frac{0.3}{0.3 + 0.2} = 0.6$$

Solved Problem 2

Suppose for 3 data points, responsibilities for cluster 1 are 0.8, 0.6, 0.4. Find N_1 .

Solution:

$$N_1 = 0.8 + 0.6 + 0.4 = 1.8$$

Solved Problem 3

Suppose data points are $x = [2, 4, 6]$ and responsibilities for one cluster are $[1, 0.5, 0.5]$. Find updated mean.

Solution:

First,

$$N_k = 1 + 0.5 + 0.5 = 2$$

Then,

$$\mu_k = \frac{1 \cdot 2 + 0.5 \cdot 4 + 0.5 \cdot 6}{2} = \frac{2 + 2 + 3}{2} = \frac{7}{2} = 3.5$$

Sequential Data

Concept

Sequential data consists of observations where order matters. Unlike standard independent data points, sequence elements are related through time or position.

A sequence may be written as:

$$X = (x_1, x_2, \dots, x_T)$$

where T is the sequence length.

Applications

- Text processing
- Speech recognition
- Sensor streams
- Financial time series

Advantages of Sequential Modeling

- Captures temporal dependence.
- Useful for forecasting and pattern tracking.
- Better represents real-world processes evolving over time.

Markov Models

Introduction

A Markov model assumes the future depends only on the present state.

Markov Property Derivation

In general, the probability of next state could depend on entire history:

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_1)$$

The first-order Markov assumption simplifies this to:

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_1) = P(X_{t+1} \mid X_t)$$

This greatly reduces modeling complexity.

Transition Probability Matrix

Define:

$$P_{ij} = P(X_{t+1} = j \mid X_t = i)$$

For all states, these probabilities form a matrix:

$$P = \begin{bmatrix} P_{11} & P_{12} & \dots \\ P_{21} & P_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Each row sums to 1.

State Update Derivation

If current state distribution is row vector $\pi^{(t)}$, then next state distribution is:

$$\pi^{(t+1)} = \pi^{(t)} P$$

This comes from weighted summation of transition probabilities from each current state.

Advantages

- Simple sequence modeling.
- Strong mathematical interpretation.
- Foundation for hidden Markov models.

Applications

- Weather modeling
- Language modeling
- User navigation modeling
- Biological sequences

Solved Problem 1

Current state vector:

$$\pi^{(0)} = [1, 0]$$

Transition matrix:

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

Find next state distribution.

Solution:

$$\pi^{(1)} = [1, 0] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = [0.7, 0.3]$$

Solved Problem 2

If current distribution is $[0.5, 0.5]$, find next state distribution using the same matrix.

Solution:

$$\pi^{(1)} = [0.5, 0.5] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

First component:

$$0.5(0.7) + 0.5(0.4) = 0.35 + 0.20 = 0.55$$

Second component:

$$0.5(0.3) + 0.5(0.6) = 0.15 + 0.30 = 0.45$$

So,

$$\pi^{(1)} = [0.55, 0.45]$$

Solved Problem 3

Using the same transition matrix, find distribution after two steps starting from $[1, 0]$.

Solution:

From Problem 1,

$$\pi^{(1)} = [0.7, 0.3]$$

Then,

$$\pi^{(2)} = [0.7, 0.3] \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

First component:

$$0.7(0.7) + 0.3(0.4) = 0.49 + 0.12 = 0.61$$

Second component:

$$0.7(0.3) + 0.3(0.6) = 0.21 + 0.18 = 0.39$$

Thus,

$$\pi^{(2)} = [0.61, 0.39]$$

Comparative Summary

Topic	Main Purpose	Key Equation	Main Strength
Probabilistic discriminative model	Predict conditional class probability	$P(y = k x) = \frac{e^{a_k}}{\sum_j e^{a_j}}$	Direct classification
Log-linear model	Linear probabilistic classification	$P(y = 1 x) = \frac{1}{1 + e^{-(w^T x + b)}}$	Interpretability

Kernel methods	Nonlinear learning	$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$	Implicit high-dimensional mapping
Sparse kernel machine	Efficient kernel prediction	$f(x) = \sum_{i \in S} \alpha_i K(x_i, x) + b$	Sparsity
Bayesian methods	Posterior inference	$P(C_k x) = \frac{P(x C_k)P(C_k)}{P(x)}$	Uncertainty modeling
k-NN	Neighbor-based classification	$d(x, x_i) = \sqrt{\sum_j (x_j - x_{ij})^2}$	Simplicity
Neural network	Nonlinear function approximation	$a = \sigma(w^T x + b)$	High flexibility
Clustering	Unsupervised grouping	$J = \sum \ x - \mu\ ^2$	Structure discovery
Mixture model	Soft density modeling	$p(x) = \sum_k \pi_k p_k(x)$	Overlapping clusters
EM algorithm	Hidden-variable estimation	E-step and M-step	Soft parameter updates
Markov model	Sequential probability transitions	$\pi^{(t+1)} = \pi^{(t)} P$	Time dependence modeling

Extra Practice Problems with Solutions

Problem 1: Logistic Probability

Given $w = [1,1]$, $x = [2,3]$, $b = -4$, find class probability.

Solution:

$$z = 1 \cdot 2 + 1 \cdot 3 - 4 = 1$$

$$P(y = 1 | x) = \frac{1}{1 + e^{-1}} \approx 0.731$$

Problem 2: Euclidean Distance

Find distance between (1,2) and (4,6).

Solution:

$$d = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25} = 5$$

Problem 3: Cluster Mean

Find the centroid of points 2,8,10.

Solution:

$$\mu = \frac{2+8+10}{3} = \frac{20}{3} \approx 6.67$$

Problem 4: Mixture Density

If $\pi_1 = 0.5$, $p_1(x) = 0.3$, $\pi_2 = 0.5$, $p_2(x) = 0.7$, find $p(x)$.

Solution:

$$p(x) = 0.5(0.3) + 0.5(0.7) = 0.15 + 0.35 = 0.5$$

Problem 5: Markov One-Step Update

Current state [0.2,0.8], transition matrix:

$$\begin{bmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{bmatrix}$$

Find next state.

Solution:

First component:

$$0.2(0.9) + 0.8(0.3) = 0.18 + 0.24 = 0.42$$

Second component:

$$0.2(0.1) + 0.8(0.7) = 0.02 + 0.56 = 0.58$$

So next state is:

$$[0.42, 0.58]$$

Final Note

This expanded sheet is intended for classroom teaching, board explanation, assignment preparation, and student revision. Students should pay special attention to how equations are derived, because derivation builds conceptual understanding and improves problem-solving ability.